Francesco Emanuele Stradi

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About Me

I am a postdoctoral researcher at Politecnico di Milano, working with Prof. Alberto Marchesi. I received my Ph.D. in computer science at Politecnico di Milano, where I was advised by Prof. Nicola Gatti. My current research focuses on the theoretical aspects of artificial intelligence, particularly online reinforcement learning (RL) and online learning. Specifically, I am interested in combining online (machine) learning theoretical tools with optimization techniques to build strategic agents able to act in complex online environments.

Experience

- **Politecnico di Milano** *May 2026 - now*
  Postdoctoral researcher Milan, Italy
  ◦ Working with Prof. Alberto Marchesi
- **lastminute.com** *June - September 2024*
  Machine Learning Scientist, internship Chiasso, Switzerland
  ◦ In charge of developing online learning algorithms for dynamic pricing
- **Polimi Artificial Intelligence Research and Innovation Center (Airic)** *August - October 2022*
  Research Scientist, internship Milan, Italy
  ◦ In charge of selecting State-Of-The-Art artificial intelligence techniques and applying them in real world business oriented scenarios

Education

- **Politecnico di Milano** *2022 - 2026*
  PhD in Computer Science Milan, Italy
  ◦ Advised by Prof. Nicola Gatti
  ◦ Conferred *cum laude*
- **Columbia University** *February - April 2025*
  Visiting Scholar New York City, USA
  ◦ Working with Prof. Christian Kroer at IEOR Department
- **Politecnico di Milano** *2019 - 2022*
  MSc in Computer Science and Engineering Milan, Italy
  ◦ Grade: 110 cum laude/110
- **Politecnico di Milano** *2016 - 2019*
  BSc in Computer Engineering Milan, Italy

Publications

C=Conference, J=Journal, W=Workshop, T=Thesis

[C.1] Kalana Kalupahana*, **Francesco Emanuele Stradi***, Matteo Castiglioni and Alberto Marchesi (2026). **Toward Optimal Regret in Robust Pricing: Decoupling Corruption and Time**. In *The 40th Conference on Neural Information Processing Systems [NeurIPS 2026].*

[C.2] Francesco Bacchiocchi*, Matteo Castiglioni*, Alberto Marchesi* and **Francesco Emanuele Stradi*** (2026). **Multi-Armed Bandits With Best-Action Queries**. In *The 40th Conference on Neural Information Processing Systems [NeurIPS 2026].*

[C.3] Eleonora Fidelia Chiefari, **Francesco Emanuele Stradi**, Matteo Castiglioni and Alberto Marchesi (2026). **Online Resource Allocation With General Constraints**. In *The 40th Conference on Neural Information Processing Systems [NeurIPS 2026].*

[C.4] Matteo Bollini*, Gianmarco Genalti*, **Francesco Emanuele Stradi***, Matteo Castiglioni and Alberto Marchesi (2026). **Replicable Constrained Bandits**. In *The 40th Conference on Neural Information Processing Systems [NeurIPS 2026].*

[C.5] **Francesco Emanuele Stradi**, Kalana Kalupahana, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2026). **Truly Adapting to Adversarial Constraints in Constrained MABs**. In *The 39th Annual Conference on Learning Theory [COLT 2026].*

[C.6] **Francesco Emanuele Stradi**, Matteo Castiglioni, Alberto Marchesi, Nicola Gatti and Christian Kroer (2025). **No-Regret Learning Under Adversarial Resource Constraints: A Spending Plan Is All You Need!**. In *The 39th Conference on Neural Information Processing Systems [NeurIPS 2025].*

[C.7] Gianmarco Genalti*, **Francesco Emanuele Stradi***, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Data-Dependent Regret Bounds for Constrained MABs**. In *The 39th Conference on Neural Information Processing Systems [NeurIPS 2025].*

[C.8] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Taming Adversarial Constraints in CMDPs**. In *The 39th Conference on Neural Information Processing Systems [NeurIPS 2025].*

[C.9] Francesco Bacchiocchi*, **Francesco Emanuele Stradi***, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Markov Persuasion Processes: Learning to Persuade From Scratch**. In *The 39th Conference on Neural Information Processing Systems [NeurIPS 2025].*

[C.10] **Francesco Emanuele Stradi**, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Learning Adversarial MDPs with Stochastic Hard Constraints**. In *The 42nd International Conference on Machine Learning [ICML 2025].*

[C.11] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Policy Optimization for CMDPs with Bandit Feedback: Learning Stochastic and Adversarial Constraints**. In *The 42nd International Conference on Machine Learning [ICML 2025].*

[C.12] **Francesco Emanuele Stradi**, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Optimal Strong Regret and Violation in Constrained MDPs via Policy Optimization**. In *The 13th International Conference on Learning Representations [ICLR 2025].*

[C.13] Davide Maran*, Francesco Bacchiocchi*, **Francesco Emanuele Stradi***, Matteo Castiglioni, Nicola Gatti and Marcello Restelli (2024). **Bandits with Ranking Feedback**. In *The 38th Conference on Neural Information Processing Systems [NeurIPS 2024].*

[C.14] **Francesco Emanuele Stradi**, Jacopo Germano, Gianmarco Genalti, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2024). **Online Learning in CMDPs: Handling Stochastic and Adversarial Constraints**. In *The 41st International Conference on Machine Learning [ICML 2024].*

[C.15] Francesco Bacchiocchi*, **Francesco Emanuele Stradi***, Matteo Papini, Alberto Maria Metelli and Nicola Gatti (2024). **Online Learning with Off-Policy Feedback in Adversarial MDPs**. In *The 33rd International Joint Conference on Artificial Intelligence [IJCAI 2024].*

[C.16] Davide Maran*, Pierricardo Olivieri*, **Francesco Emanuele Stradi***, Nicola Gatti and Marcello Restelli (2024). **Online Markov Decision Processes Configuration with Continuous Decision Space**. In *The 38th AAAI Conference on Artificial Intelligence [AAAI 2024].*

[J.1] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2026). **Policy Optimization for CMDPs with Bandit Feedback: Best-of-Both-Worlds and Beyond**. In *Artificial Intelligence [AIJ 2026].*

[W.1] Eleonora Fidelia Chiefari, **Francesco Emanuele Stradi**, Matteo Castiglioni and Alberto Marchesi (2026). **Handling General Constraints in Online Resource Allocation**. In *The 19th European Workshop on Reinforcement Learning [EWRL 2026].*

[W.2] Gianmarco Genalti*, **Francesco Emanuele Stradi***, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Data-Dependent Regret Bounds for MABs with Constraints**. In *NeurIPS 2025 Workshop: Aligning Reinforcement Learning Experimentalists and Theorists [ARLET 2025].*

[W.3] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Policy Optimization in CMDPs with Bandit Feedback: Learning with Stochastic and Adversarial Constraints**. In *NeurIPS 2025 Workshop: Aligning Reinforcement Learning Experimentalists and Theorists [ARLET 2025].*

[W.4] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Taming Adversarial Constraints in Constrained MDPs**. In *The 18th European Workshop on Reinforcement Learning [EWRL 2025].*

[W.5] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Policy Optimization for CMDPs with Bandit Feedback: Learning Stochastic and Adversarial Constraints**. In *The 18th European Workshop on Reinforcement Learning [EWRL 2025].*

[W.6] **Francesco Emanuele Stradi**, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2024). **Online learning in CMDPs with adversarial losses and stochastic hard constraints**. In *The 17th European Workshop on Reinforcement Learning [EWRL 2024].*

[W.7] Francesco Bacchiocchi*, **Francesco Emanuele Stradi***, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2024). **Markov Persuasion Processes: How to Persuade Multiple Agents From Scratch**. In *ICML 2024 Workshop: Aligning Reinforcement Learning Experimentalists and Theorists [ARLET 2024].*

[W.8] Francesco Bacchiocchi*, **Francesco Emanuele Stradi***, Matteo Papini, Alberto Maria Metelli and Nicola Gatti (2023). **Online Adversarial MDPs with Off-Policy Feedback and Known Transitions**. In *The 16th European Workshop on Reinforcement Learning [EWRL 2023].*

[W.9] Davide Maran*, Pierricardo Olivieri*, **Francesco Emanuele Stradi***, Nicola Gatti and Marcello Restelli. (2023). **Online Configuration in Continuous Decision Space**. In *The 16th European Workshop on Reinforcement Learning [EWRL 2023].*

[T] **Francesco Emanuele Stradi** (2026). **Regret Minimization in Constrained Markov Decision Processes**. External Reviewers: Prof. Haipeng Luo, Matteo Pirotta, PhD.

* stands for equal contribution

Teaching

- **Probabilità e Statistica Matematica, TA** *Spring 2024*
  Politecnico di Milano
  - BSc in Management Engineering
- **Statistica, TA** *Spring 2024*
  Politecnico di Milano
  - BSc in Mechanical Engineering

Student Mentoring and Co-Supervision

- **PhD Students:**
  - **Maria Elena Vischi** (2026–present). Research mentoring on *Online Market Making*
  - **Eleonora Fidelia Chiefari** (2025–present). Research mentoring on *Online Resource Allocation*
- **MSc Thesis Co-Supervision:**
  - **Kalana Kalupahana** (2026). *Constrained bandit learning under time-varying stochastic constraints*
  - **Emanuele Pocelli** (2026). *A primal-dual approach for online learning in constrained tree-form sequential decision problem*
  - **Eleonora Fidelia Chiefari** (2025). *Online learning in stochastic and adversarial constrained Markov decision processes*
  - **Matteo Badialetti** (2025). *Constrained Contextual Markov Decision Processes*
  - **Anna Lunghi** (2024). *Dealing with non-stationarity in Constrained Markov Decision Processes*
  - **Giuseppe Urso** (2023). *Online Markov Decision Processes configuration with continuous decision space*
  - **Filippo Cipriani** (2023). *Dynamic pricing in multi-state environments*

Service

- **Reviewing:** ICLR (2025, 2026, 2027), ICML (2025, 2026 **Silver Reviewer**), NeurIPS (2024, 2025, 2026), Mathematics of Operations Research (2025), Transactions on Machine Learning Research (2026), ITCS (2027), Aistats (2027)
- **Program Committee:** EC (2026)

Programming

- **Programming Languages:** C, Java, Python, SQL, Ampl

Additional Information

- **Languages:** Italian (Native), English (Proficiency level)

References

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